INCH-POUND

MIL-PRF-20/36F
21 June 2001
SUPERSEDING
MIL-PRF-20/36E
6 May 1988

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR06 AND CC06

Style CC06 is inactive for new design after 19 September 1983. Use CCR06.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-PRF-20.

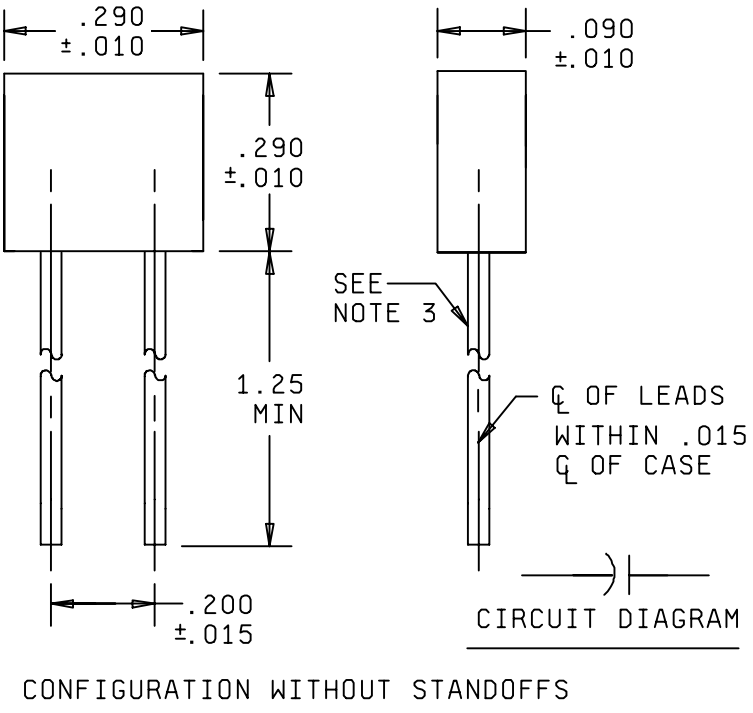

FIGURE 1. Styles CCR06 and CC06 capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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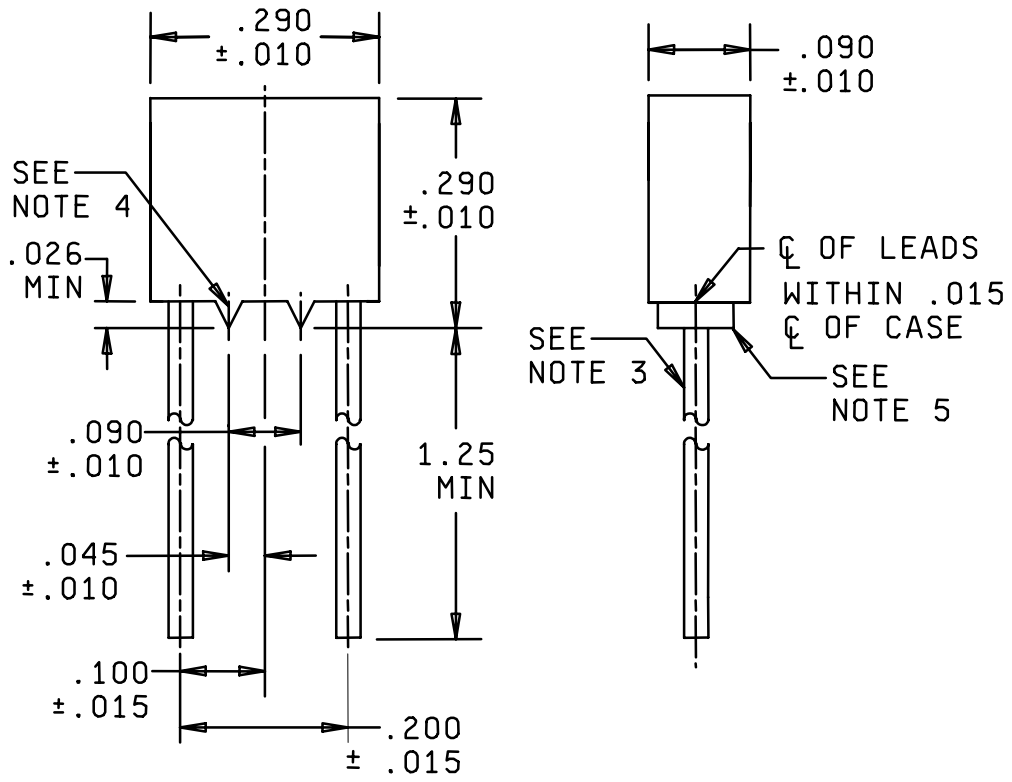

OPTIONAL CONFIGURATION WITH STANDOFFS (CCR06 ONLY)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Lead diameter shall be .023 (0.58 mm) to .029 (0.74 mm).
4. Optional standoffs for style CCR06 shall be of triangular shape so that the standoffs will provide line contact with surface upon which the capacitor is mounted. The shape of the triangle is optional.
5. Thickness of standoffs shall be .075 (1.90 mm) to the body thickness.
6. Lead length may be a minimum of .625 (15.88 mm) for use in tape and reel automatic insertion equipment, when specified.
7. At the option of the user, the standoff configuration may be furnished as a replacement for the nonstandoff configuration of the same style.

Inches	mm
.010	0.25
.015	0.38
.026	0.66
.045	1.14
.090	2.29
.100	2.54
.200	5.08
.290	7.37
1.250	31.80

FIGURE 1. Styles CCR06 and CC06 capacitors - Continued.

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REQUIREMENTS:

Dimensions and configuration: see figure 1.

Lead type: Radial.

Case type: Molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

Failure rate level (CCR06 only): M (1.0 percent), P (0.1 percent), R (.01 percent) or S (.001 percent).

Thermal shock and voltage conditioning (CCR06 only): In accordance with MIL-PRF-20.

TABLE I. Capacitor characteristics.

Type designation 1/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance	Type designation 1/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance
CC-06CG361---	200	360	FGJ	CC-06CG242---	100	2,400	FGJ
CC-06CG391---	200	390	FGJ	CC-06CG272---	100	2,700	FGJ
CC-06CG431---	200	430	FGJ	CC-06CG302---	100	3,000	FGJ
CC-06CG471---	200	470	FGJ	CC-06CG332---	100	3,300	FGJ
CC-06CG511---	200	510	FGJ	CC-06CG362---	100	3,600	FGJ
CC-06CG561---	200	560	FGJ	CC-06CG392---	100	3,900	FGJ
CC-06CG621---	200	620	FGJ	CC-06CG432---	100	4,300	FGJ
CC-06CG681---	200	680	FGJ	CC-06CG472---	100	4,700	FGJ
CC-06CG751---	200	750	FGJ	CC-06CG512---	50	5,100	FGJK
CC-06CG821---	200	820	FGJ	CC-06CG562---	50	5,600	FGJK
CC-06CG911---	200	910	FGJ	CC-06CG622---	50	6,200	FGJK
CC-06CG102---	200	1,000	FGJ	CC-06CG682---	50	6,800	FGJK
CC-06CG112---	200	1,100	FGJ	CC-06CG752---	50	7,500	FGJK
CC-06CG122---	200	1,200	FGJ	CC-06CG822---	50	8,200	FGJK
CC-06CG132---	200	1,300	FGJ	CC-06CG912---	50	9,100	FGJK
CC-06CG152---	200	1,500	FGJ	CC-06CG103---	50	10,000	FGJK
CC-06CG162---	200	1,600	FGJ	CC-06CG123---	50	12,000	FGJK
CC-06CG182---	200	1,800	FGJ	CC-06CG153---	50	15,000	FGJK
CC-06CG202---	100	2,000	FGJ	CC-06CG183---	50	18,000	FGJK
CC-06CG222---	100	2,200	FGJ				

1/ Complete type designation will include the following:

1st dash - Symbol “R” (for style CCR06) or dash will be deleted (for style CC06).

2nd dash - Applicable capacitance tolerance symbol.

3rd dash - Applicable failure rate level symbol (CCR06 only) or dash will be deleted (for style CC06).

4th dash - Symbol “V” for standoffs CCR06 only) or dash will be deleted (for style CCR06 without standoffs and style CC06).

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Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-PRF-20.

Dielectric withstanding voltage: In accordance with MIL-PRF-20.

Body insulation: Test II.

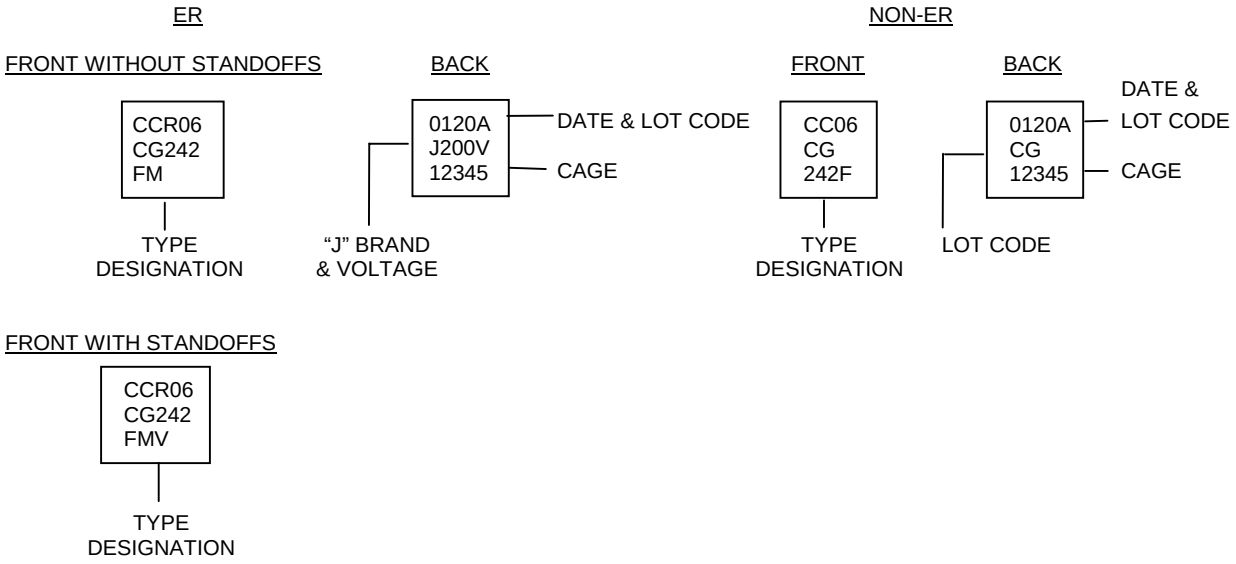
Solderability: The leads shall be solderable up to .020 inch (0.51 mm) from the body egress.

Resistance to soldering heat: In accordance with MIL-PRF-20.

Workmanship: For style CCR06 with standoff, exposed copper or bare leads shall be permitted to a maximum of .020 inch (0.51 mm) from the body egress. For style CCR06 without standoffs and style CC06, workmanship shall be in accordance with MIL-PRF-20.

Life: In accordance with MIL-PRF-20, operating condition 2.

Marking: Method I of MIL-STD-1285. At the option of the manufacturer, the marking may be placed on one side of the capacitor, or the marking may be placed on two lines as long as it is in the same order as shown in the following examples:



Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - AS, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

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